



UNSUPERVISED LEARNING

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Unsupervised machine-learning

- Unsupervised learning is a branch of machine learning that works with unlabelled data to discover hidden patterns and structures.
- Unlike supervised learning, it does not rely on predefined outcomes, allowing the model to explore relationships autonomously.
- It is especially useful for complex healthcare data where manual labelling is difficult or impossible.
- However, evaluating and interpreting unsupervised models can be challenging and often requires domain expertise.

Unsupervised machine-learning

Clustering

- Grouping similar data points together

Dimensionality Reduction

- Reducing features while preserving information

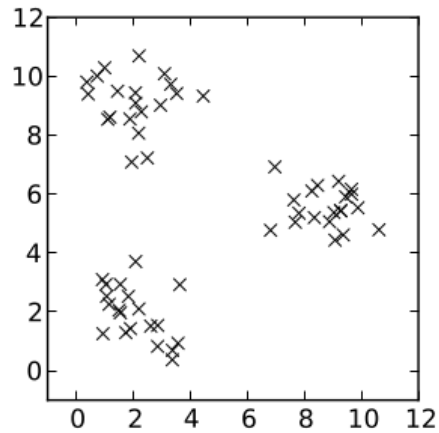
Association Rule Learning

- Discovering relationships between variables

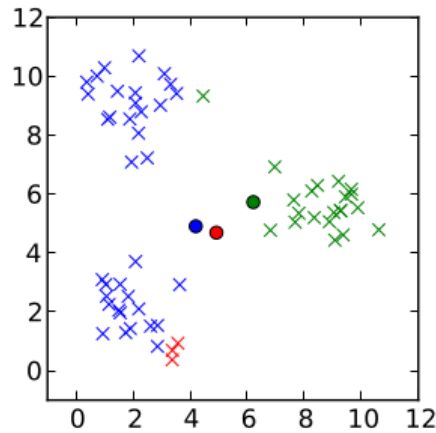
Anomaly Detection

- Identifying unusual patterns

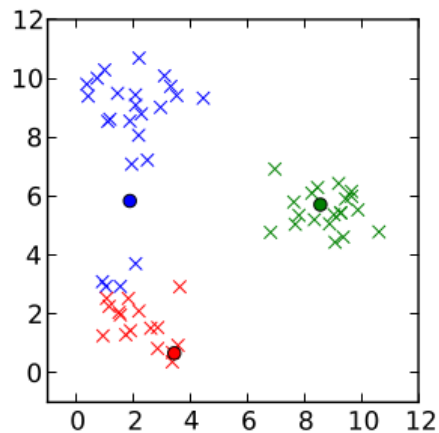
k -Means



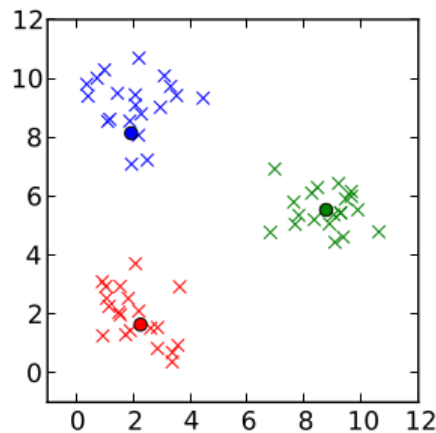
(a) dataset.



(b) step 1.



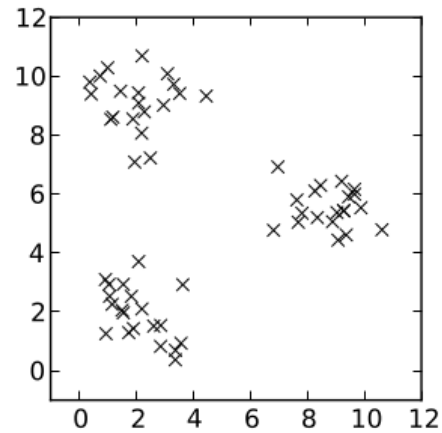
(c) step 2.



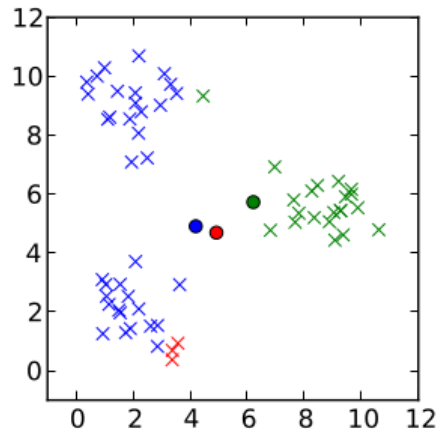
(d) step 3.

- k -means assumes there are k clusters, and each point is close to its cluster center (the mean of points in the cluster).
- If we knew the cluster assignment we could easily compute means.
- If we knew the means we could easily compute cluster assignment.
- Chicken and egg problem!

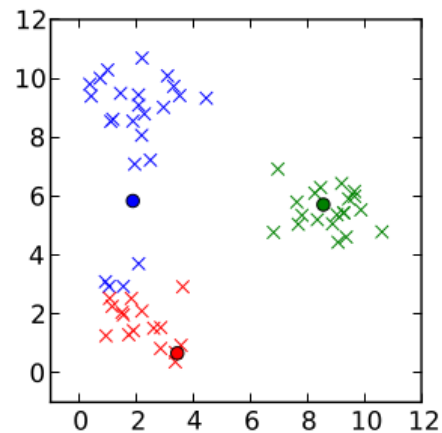
k-Means



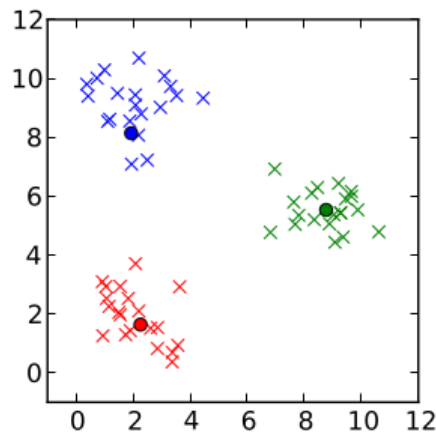
(a) dataset.



(b) step 1.



(c) step 2.



(d) step 3.

- Randomly initialize cluster centers
- The algorithm iteratively alternates between two steps:
 - Assignment step: Assign each data point to the closest cluster
 - Refitting step: Move each cluster center to the center of gravity of the data assigned to it
- Each cluster is represented by its centroid (mean), and the algorithm tries to minimize within-cluster variance to make points as close as possible to their cluster center
- Output
 - Cluster labels (which group each observation belongs to)
 - Cluster centroids (means)

k-Means

Advantages

- Extremely fast and scalable
- Simple to understand and implement
- Works well with large datasets
- Computationally efficient

Disadvantages

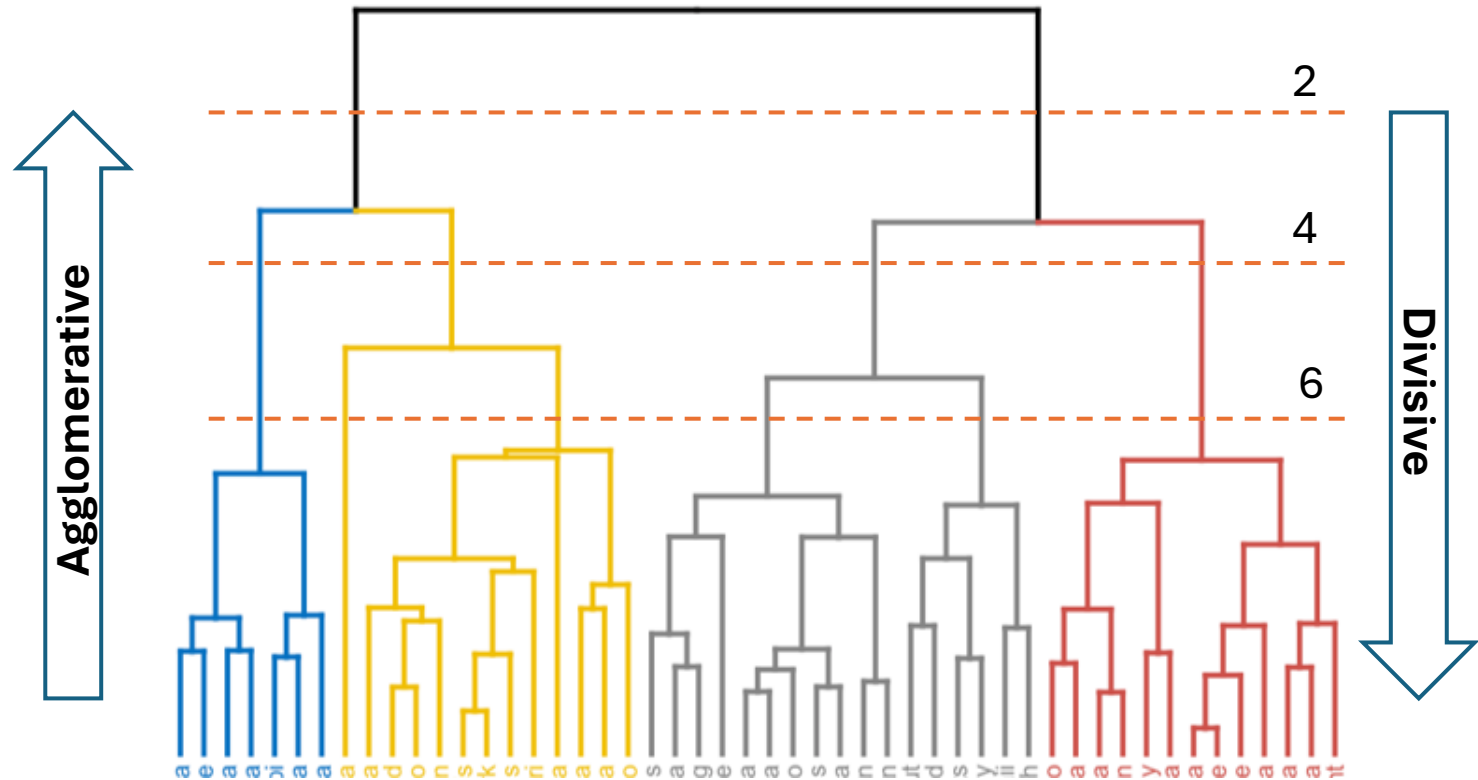
- Sensitive to initial seed placement
- Assumes spherical clusters of similar size
- Requires specifying k in advance
- Poor performance on non-convex clusters

When to use

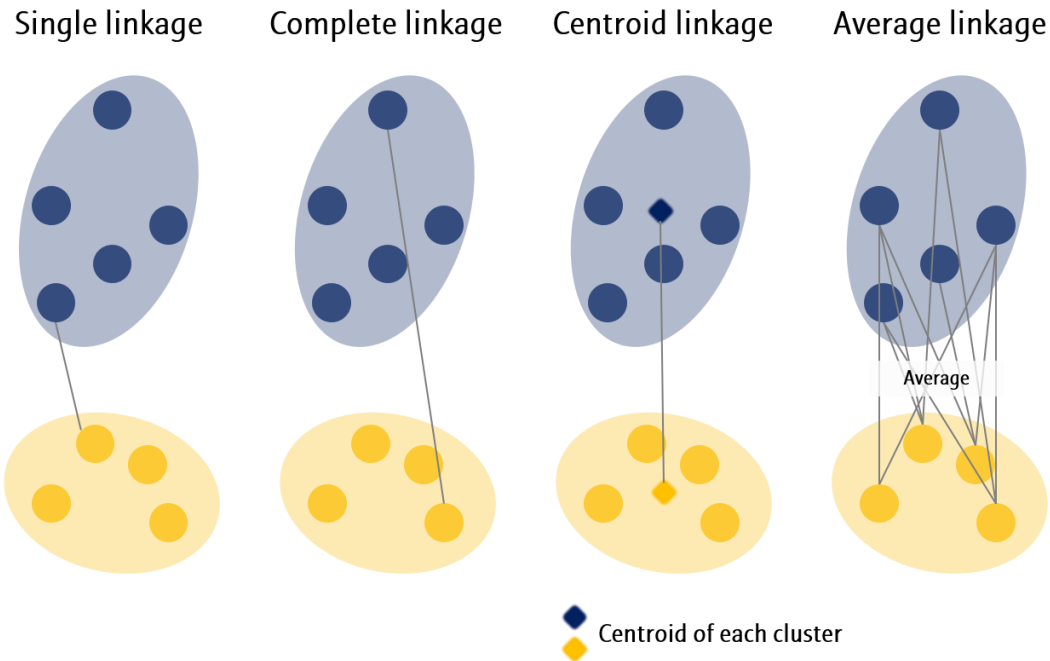
- Large datasets where speed matters, customer segmentation, image compression, or when you have a reasonable estimate of cluster count.
- Research analyzing k -means found it to be among the most widely applied unsupervised algorithms, with consistent performance across diverse applications.

Hierarchical clustering

- Hierarchical clustering builds a tree-like structure of clusters by either
 - iteratively merging the closest clusters (agglomerative) or
 - recursively splitting a single cluster into smaller ones (divisive),based on pairwise distances.
- It does not require specifying the number of clusters in advance but is computationally more intensive than methods like *k*-means.



Hierarchical clustering



Distance between two clusters:

- **Single linkage**
 - Shortest distance between two clusters, A and B.
- **Complete linkage**
 - Farthest distance between two clusters
- **Average linkage**
 - Average of distances between all pair of points in the two clusters, A and B.
- **Centroid Linkage**
 - Distance between the centroids (mean) of clusters A and B.

Hierarchical clustering

Advantages

- No need to specify k in advance
- Produces interpretable dendrograms
- Multiple cutting heights give different granularities
- Works with various distance metrics

Disadvantages

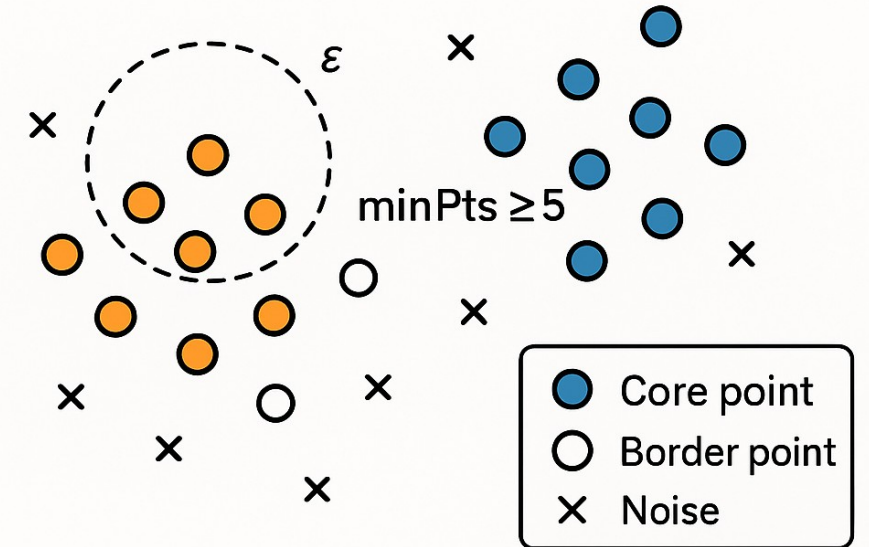
- Time complexity – slow on large datasets
- Sensitive to noise and outliers
- Difficult to determine optimal cut level
- Once merged, cannot undo (greedy approach)

When to use

- Small to medium datasets, exploratory analysis, when cluster hierarchy is meaningful (e.g., taxonomy), or visualization needs.

DBSCAN

- DBSCAN (Density-Based Spatial Clustering of Applications with Noise) groups data points based on density, identifying clusters as areas where many points are close together while treating sparse regions as noise.
- It works by classifying points into
 - core points (with at least MinPts neighbours within distance ϵ),
 - border points (near core points but with fewer neighbours), and
 - noise points (isolated points not belonging to any cluster),allowing it to detect clusters of arbitrary shape without needing to predefine the number of clusters.



DBSCAN

Advantages

- Discovers clusters of arbitrary shapes
- Automatically detects outliers/noise
- No need to specify cluster count
- Robust to outliers
- Can find clusters of varying sizes

Disadvantages

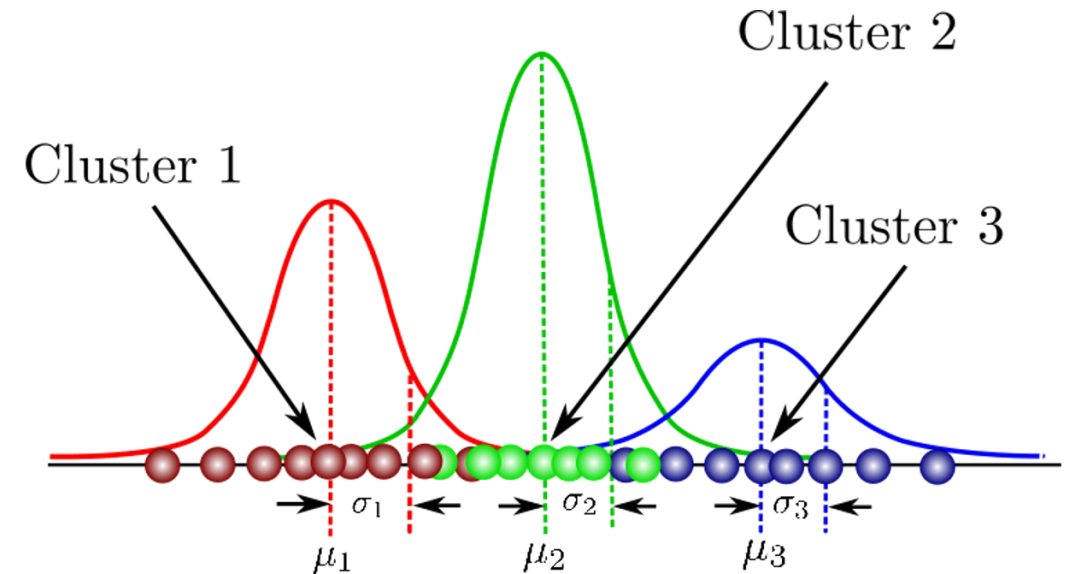
- Struggles with varying density (dense vs. sparse) clusters
- Parameter selection (eps) can be challenging
- Performance degrades in high dimensions

When to use

- Spatial data, when cluster count is unknown, data contains noise/outliers, or non-spherical cluster shapes.

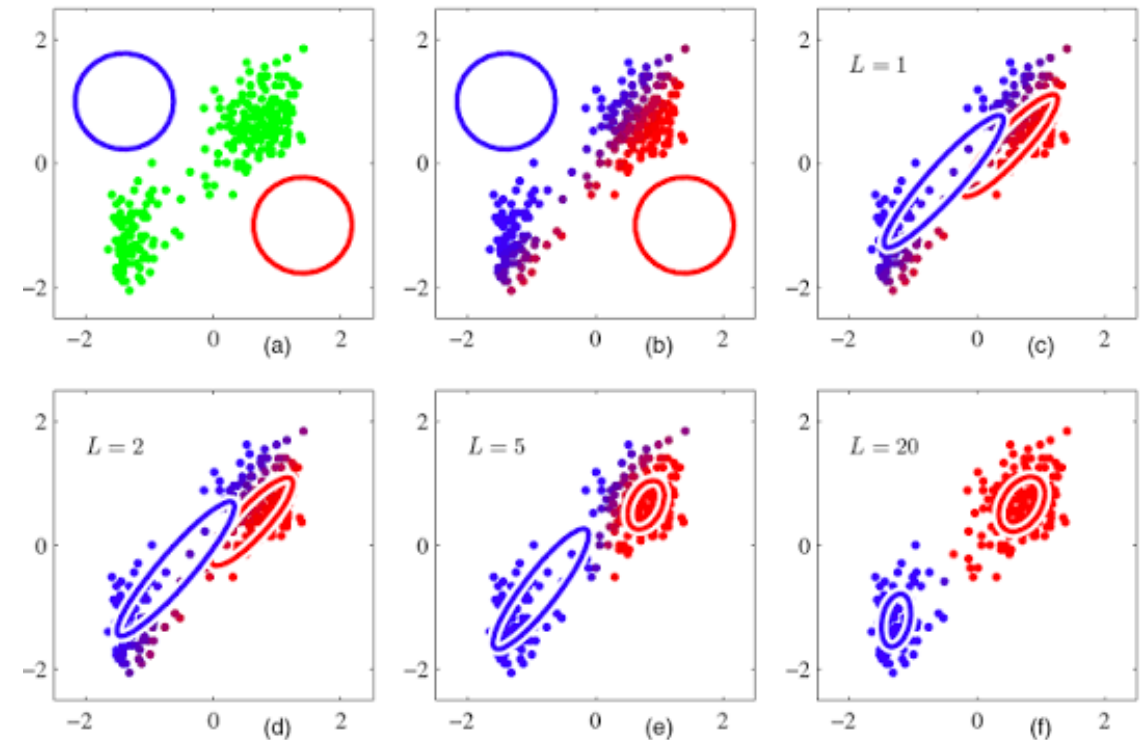
Gaussian Mixture Models

- Probabilistic model that assumes all the data points are generated from a weighted mixture of a finite number of Gaussian distributions with unknown parameters. One can think of mixture models as generalizing *k*-means clustering to incorporate information about the covariance structure of the data as well as the centers of the latent Gaussians.



Gaussian Mixture Models

- The Expectation–Maximization (EM) algorithm for Gaussian Mixture Models (GMMs) is an iterative method used to estimate model parameters when the true cluster labels (which Gaussian each point belongs to) are unknown.
 - Step 1 (Expectation step): Assign each data point a probability of belonging to each Gaussian component
 - Step 2 (Maximization step): Update the parameters (means, variances, weights) to best fit those assignments
- This process is repeated until convergence, gradually improving the model and reaching a local optimum of the data likelihood.



Gaussian Mixture Models

Advantages

- Provides probabilistic cluster assignments
- Can capture elliptical cluster shapes
- Flexible covariance structures
- Theoretically well-founded

Disadvantages

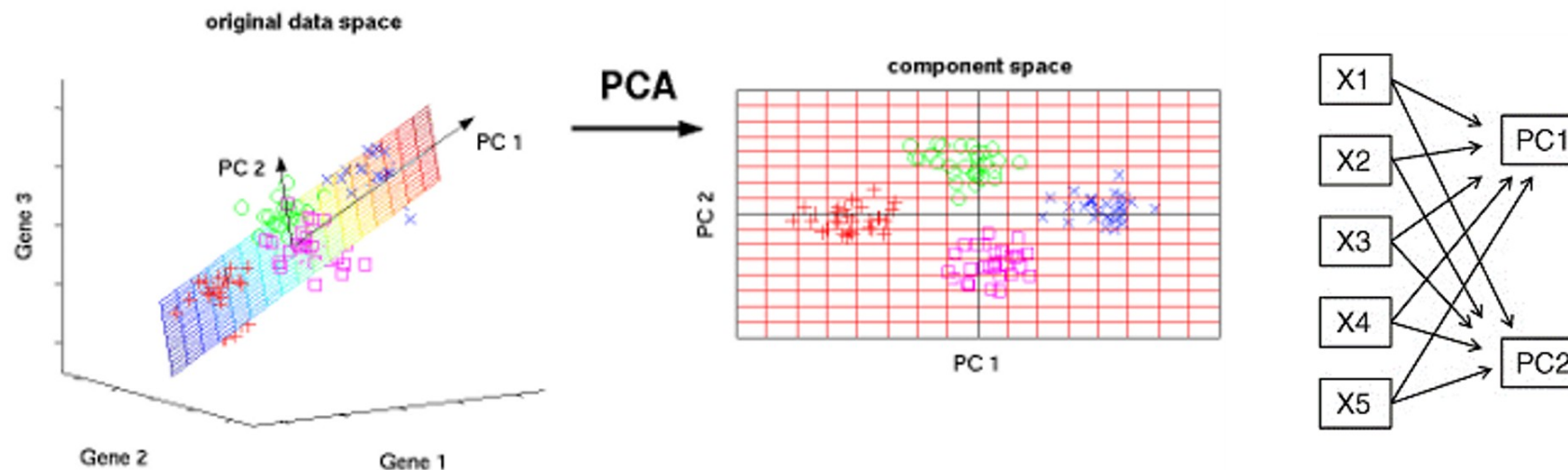
- Sensitive to initialization
- Computationally expensive in high dimensions
- Can converge to local optima
- Requires careful selection of component count

When to use

- When you need probability estimates for cluster membership, data has elliptical structures, or when modeling complex distributions.

Principal Component Analysis

- Transforms the original data into a new coordinate system.
- PCA is effective in reducing dimensionality while preserving the most significant variance in the dataset.
- Principal components (PCs) are extracted sequentially, with each component capturing the maximum remaining variance while remaining orthogonal (uncorrelated) to the previous ones, thereby removing redundancy in the data.
- Selecting the top k components reduces the data to k dimensions while preserving most of its important variation.



Principal Component Analysis

Advantages

- Fast and computationally efficient
- Good for visualization (2D/3D projections)
- Reduces noise and redundancy
- Improves downstream algorithm performance
- Interpretable components (linear combinations)

Disadvantages

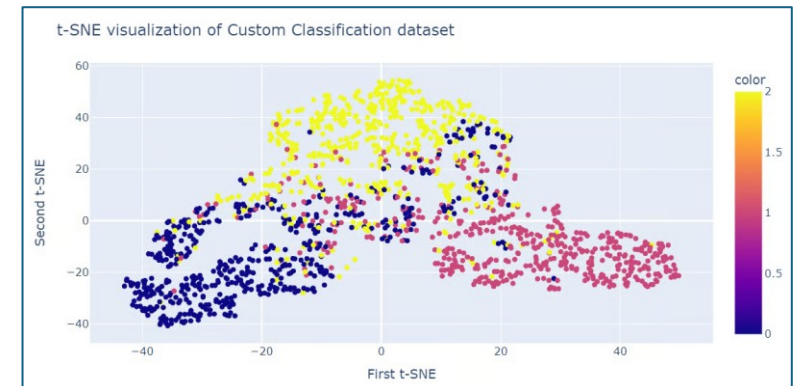
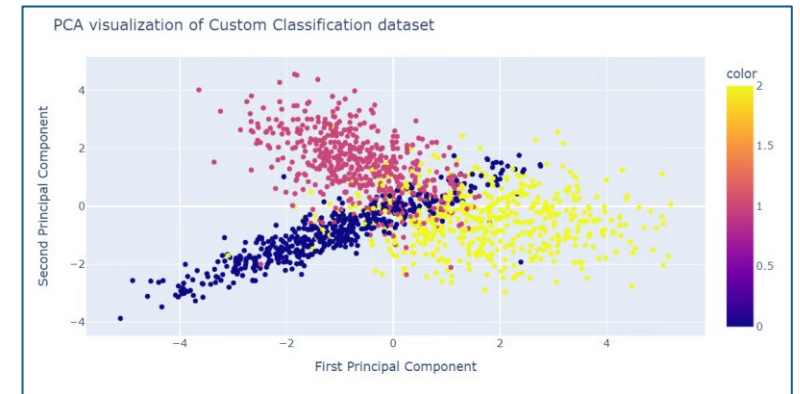
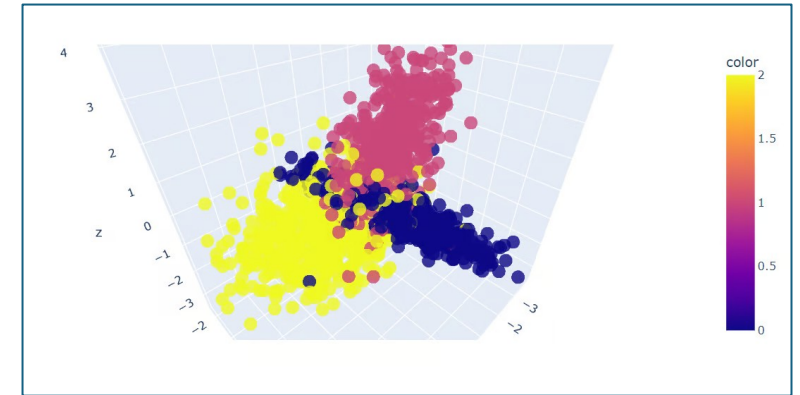
- Only captures linear relationships
- Sensitive to feature scaling
- Hard to interpret in some domains
- Information loss with compression
- Assumes Gaussian distribution

When to use

- High-dimensional data visualization, feature reduction before modeling, noise reduction, or when linear relationships dominate.

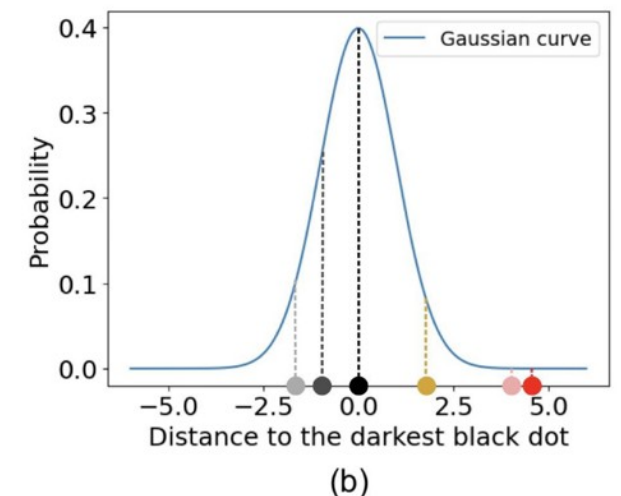
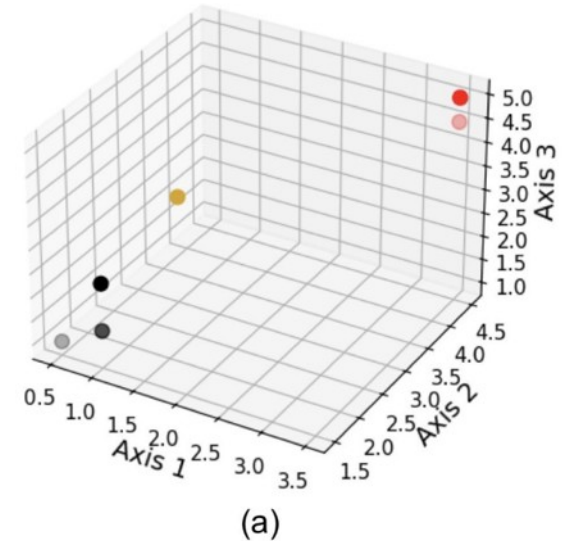
t-SNE

- PCA is a linear dimensionality reduction method that preserves overall variance and large distances between data points, making it suitable for structured data with linear relationships.
- In contrast, t-SNE (t-distributed Stochastic Neighbor Embedding) is a nonlinear (allows us to separate data that cannot be separated by a straight line) technique that preserves local similarities between points, making it better for visualizing complex patterns and clusters in high-dimensional data.
- t-SNE gives you a feel and intuition on how data is arranged in higher dimensions. It is often used to visualize complex datasets into two and three dimensions, allowing us to understand more about underlying patterns and relationships in the data.



t-SNE

- t-SNE starts by calculating a pairwise similarity between all data points in the high-dimensional space using a Gaussian kernel. The points far apart have a lower probability of being picked than the points close together.
- The algorithm then tries to map higher-dimensional data points onto lower-dimensional space while preserving the pairwise similarities.
- It is achieved by minimizing the divergence between the original high-dimensional and lower-dimensional probability distribution. The algorithm uses gradient descent to minimize the divergence. The lower-dimensional embedding is optimized to a stable state.
- The optimization process allows the creation of clusters and sub-clusters of similar data points in the lower-dimensional space, which are visualized to understand the structure and relationships in the higher-dimensional data.



t-SNE

Advantages

- Excellent for visualization
- Reveals local structure
- Handles complex manifolds

Disadvantages

- Not for dimensionality reduction (only visualization)
- Computationally expensive and sensitive to hyperparameters
- Non-deterministic results and lack of interpretability
- Not suitable for out-of-sample data

When to use

- Visualizing high-dimensional data (e.g., word embeddings, image features), exploratory analysis

UMAP

- In recent years, UMAP (Uniform Manifold Approximation and Projection) has emerged as a popular alternative to t-SNE.
- While both are non-linear dimensionality reduction techniques designed for visualization, UMAP addresses some of the limitations of t-SNE:

UMAP is faster and more memory-efficient than t-SNE, making it suitable for large datasets.

UMAP is better at preserving both local and global structures in the data, while t-SNE primarily focuses on preserving local pairwise similarities.

UMAP has fewer hyperparameters to tune compared to t-SNE, and its results are generally more consistent across runs.

Unlike t-SNE, UMAP supports transformations for new data points, making it more suitable for dynamic datasets.

Evaluation metrics

Internal Metrics (No labels needed)

- Silhouette Score (0.6–0.9 = good): Measures cluster cohesion and separation
- Davies-Bouldin Index (lower = better): Ratio of within-cluster to between-cluster distances
- Calinski-Harabasz Index (higher = better): Ratio of between-cluster to within-cluster variance

External Metrics (Requires ground truth)

- Adjusted Rand Index (ARI) (>0.9 excellent, >0.8 good): Measures agreement with true labels
- Normalized Mutual Information (NMI): Information shared between predicted and true clusters

Best Practices

- Use multiple metrics for comprehensive evaluation
- Visual inspection remains crucial
- Consider domain-specific knowledge and metrics

Example

Original Research

Clustering of cardiovascular risk factors and their association with socio-demographic and lifestyle factors in middle-aged and older adults in rural northeast South Africa: Findings from HAALSI

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Mhembere, M.N. et al. (2026) 'Clustering of cardiovascular risk factors and their association with socio-demographic and lifestyle factors in middle-aged and older adults in rural northeast South Africa: Findings from HAALSI', *Public Health*, 254, p. 106246. Available at: <https://doi.org/10.1016/j.puhe.2026.106246>.

Problem definition

- This study addresses the growing burden of cardiovascular disease (CVD) in rural South Africa.
- Key issue: risk factors (e.g., obesity, hypertension, high cholesterol) often co-occur rather than act independently.
- Objective
 - Identify natural clusters of cardiovascular risk factors
 - Determine how these clusters relate to socio-demographic and lifestyle factors

Dataset

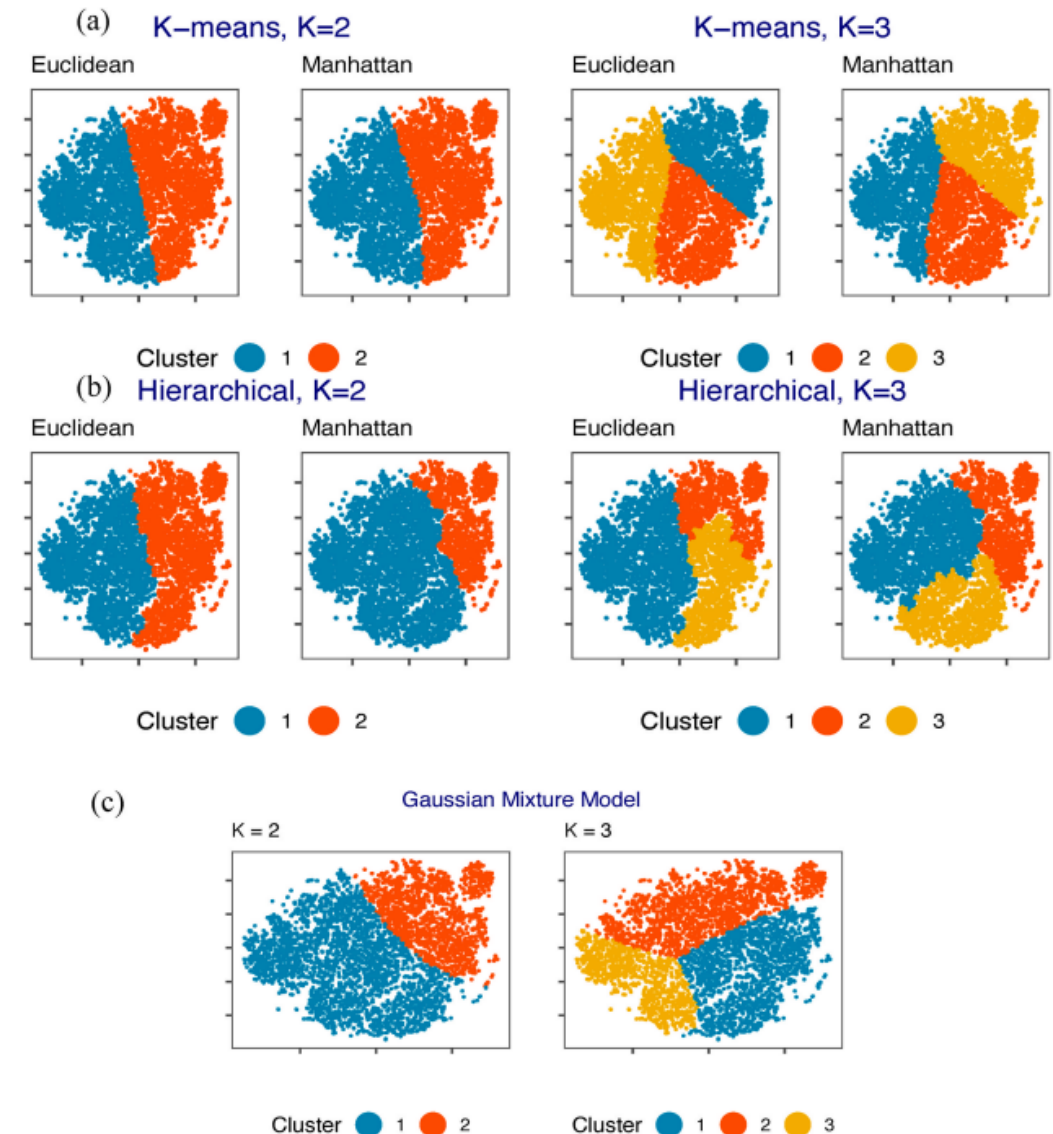
- Source: HAALSI cohort (Health and Aging in Africa study)
- Sample size: 5059 individuals aged 40+ years
- Data types included:
 - Clinical/biomarker data: BMI, blood pressure, glucose, cholesterol, triglycerides
 - Anthropometric measures: Waist-to-hip ratio, weight, height
 - Lifestyle data: Smoking, alcohol use, diet, physical activity
 - Socio-demographic variables: Age, sex, education, wealth status

Data pre-processing

- Handling missing data
 - Used Multiple Imputation (MICE) to estimate missing values
- Feature standardisation
 - Clinical thresholds applied (e.g., hypertension ≥ 140 mmHg, glucose ≥ 7 mmol/L)
- Feature construction
 - Derived variables such as: BMI categories, waist-to-hip ratio (WHR), physical activity (MET-minutes/week)
- Cluster preparation
 - Selected key biomarkers for clustering
 - Ensured variables were comparable and interpretable

ML model development

- Unsupervised learning (core model)
- Three clustering algorithms were applied:
 - K-means clustering
 - Hierarchical clustering
 - Gaussian Mixture Models
- These methods grouped individuals based on similar cardiovascular risk profiles
- Optimal number of clusters determined using:
 - Elbow method
 - Silhouette score
- Result: Identified 2 main clusters:
 - Cluster 1 → low-risk (optimal biomarkers)
 - Cluster 2 → high-risk (elevated biomarkers)



ML Model development

- Supervised learning (explanatory model)
- Multivariable logistic regression used after clustering:
 - Model probability of belonging to high-risk cluster
 - Identify associations with age, sex, education, wealth, lifestyle factors (smoking, alcohol, diet, physical activity)
- This is a hybrid ML framework:
 - Unsupervised used for discovery
 - Supervised used for explanation

Evaluation

- Cluster evaluation using internal validation:
 - Silhouette score
- Compared consistency across:
 - K-means
 - Hierarchical clustering
 - GMM
- Logistic regression used:
 - Identified significant predictors of high-risk cluster membership
- Ensured clusters were stable and meaningful

Example 2

Population-based clustering of co-occurring social determinants: An application of unsupervised machine learning

Ingrid Giesinger^a , Emmalin Buajitti^a, Arjumand Siddiqi^{a,b}, Peter M. Smith^{a,c},
Rahul G. Krishnan^{d,e,f}, Gemma Postill^a, Laura C. Rosella^{a,e,g,h,*}

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Giesinger, I. et al. (2026) 'Population-based clustering of co-occurring social determinants: An application of unsupervised machine learning', *Annals of Epidemiology*, 117, p. 110062. Available at: <https://doi.org/10.1016/j.annepidem.2026.110062>.

Problem definition

- This study addresses the challenge of understanding complex, co-occurring social determinants of health.
- Traditional methods analyze these factors individually or as simple composite scores, which may miss real-world complexity.
- Objective
 - Use unsupervised machine learning to identify data-driven population subgroups based on multiple social exposures
 - Link these subgroups to health outcomes (premature mortality)

Dataset

- Large population-based cohort from Ontario (Canada)
 - Canadian Community Health Survey (CCHS: 2001–2012)
- Linked with:
 - Census data
 - Vital statistics (mortality records)
- Sample size: ~115,000 individuals (age 25–64)
- Data types:
 - Individual-level variables: Income, education, employment, immigration status, food insecurity
 - Area-level variables: Neighbourhood deprivation, housing, population characteristics

Data pre-processing

- Handling scale and outliers
 - Continuous variables winsorized (1st–99th percentile) to reduce outliers
 - Standardised to ensure comparability across variables
- Addressing multicollinearity
 - Highly correlated variables removed: Pearson's $r > 0.9$ (continuous); Cramér's $V > 0.9$ (categorical)
- Dimensionality reduction
 - Applied PCA
 - Retained components explaining ~90% of variance
- Feature structuring
 - Combined individual-level + area-level social determinants e.g. assigning census-based measures using individual's postal code

ML model development

- Explored multiple algorithms (partitioning, density-based, model-based) using an OPTICS plot to assess the best algorithm
- Tested cluster sizes: $k = 2$ to $k = 12$
- Evaluated using:
 - Silhouette score
 - Davies–Bouldin index
 - Cluster cost
 - Cohesion

ML model development

- Selected model: K-prototypes clustering
 - Hybrid of: K-means (continuous data) and K-modes (categorical data)
- Chosen because:
 - Handles mixed data types
 - Scales to large datasets
 - Produces interpretable clusters
- Two solutions:
 - 4-cluster solution → best statistical fit
 - 6-cluster solution → better interpretability and granularity
- Identified distinct social profiles:
 - Economically marginalised
 - Immigrant-dense disadvantaged groups
 - Socioeconomically secure groups
 - Rural vs urban variations

Evaluation

- Internal validation: Cluster quality assessed using
 - Silhouette score
 - Davies–Bouldin index
 - Cluster stability (Cramér's $V > 0.65$ across runs)
- Visual validation: Used dimensionality reduction methods PCA, t-SNE and UMAP to verify the separation and structure of clusters

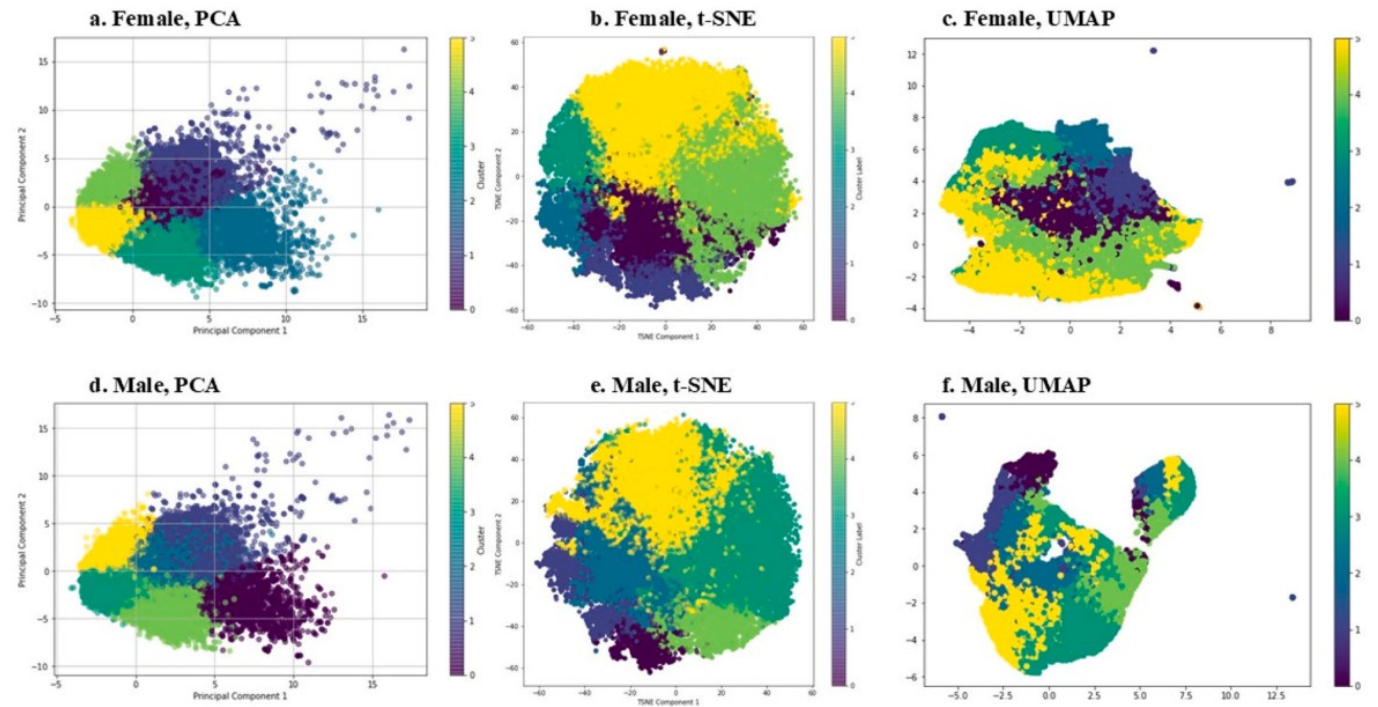


Fig. 2. Visualisation of sex-stratified clustering patterns across three dimensionality reduction methods, 6-cluster solution.

Conclusion

- Machine learning has the potential to transform healthcare systems. It enables:
 - More accurate and timely diagnosis
 - Identification of patterns and trends in patient data
 - Streamlined administrative processes
 - Personalised treatment and care pathways
- Successful integration requires:
 - Deep understanding of the healthcare context
 - Strong collaboration between clinicians and data scientists
 - Commitment to ethical, responsible, and patient-centred models

